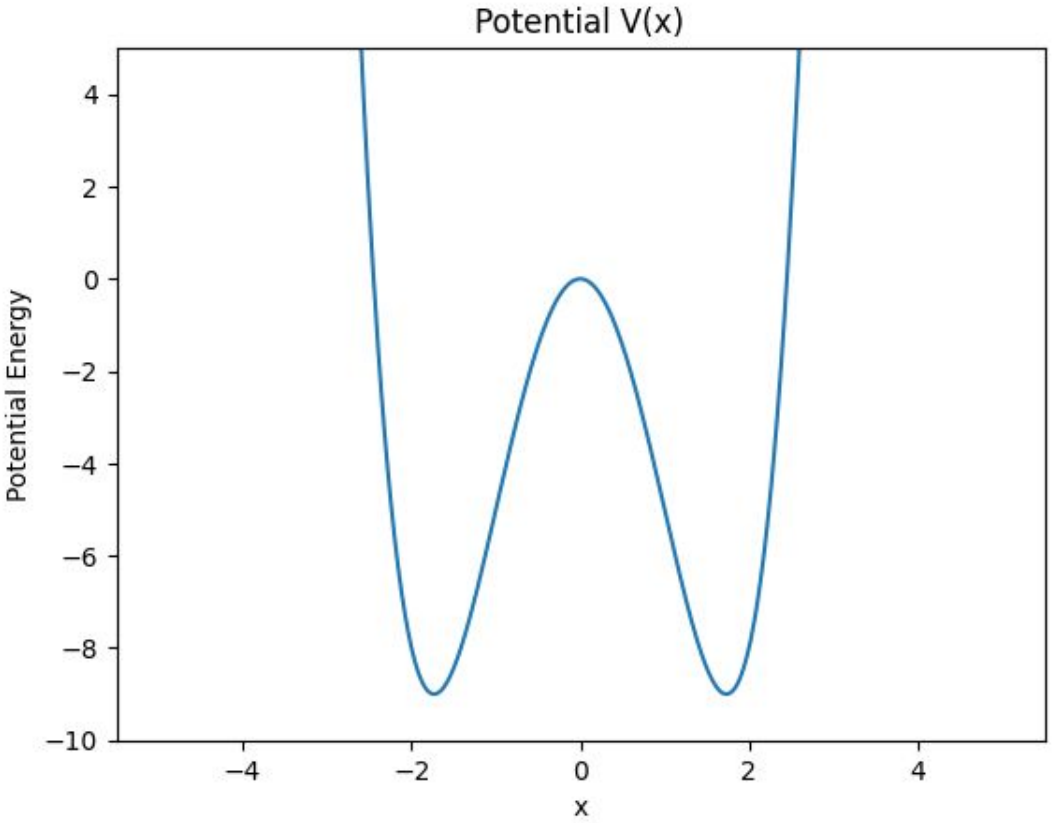
Double-well potential

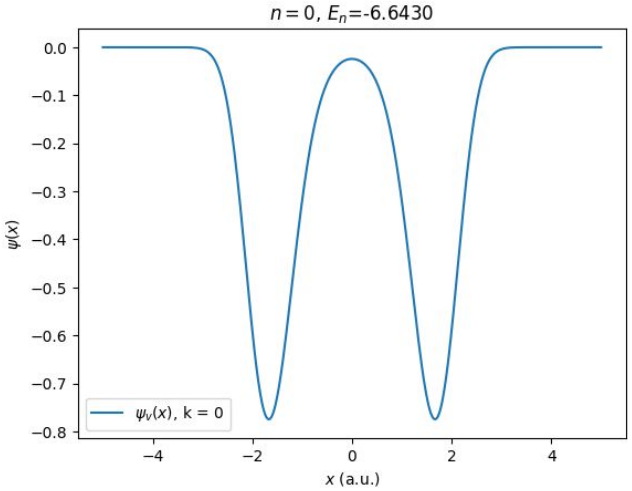


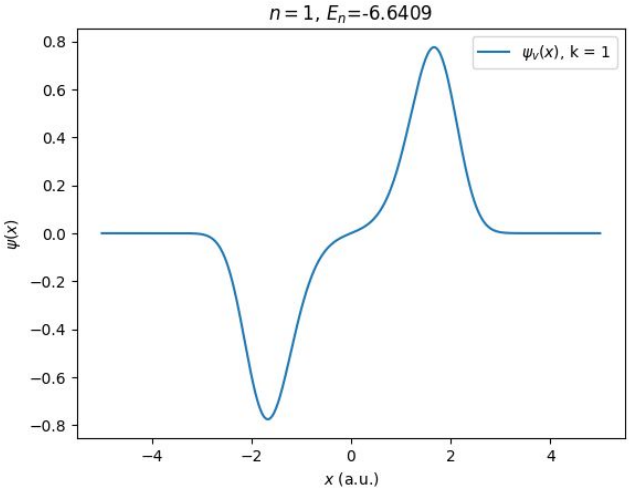
$\mathbf{V(x) = b \cdot x^4 - c \cdot x^2}$,

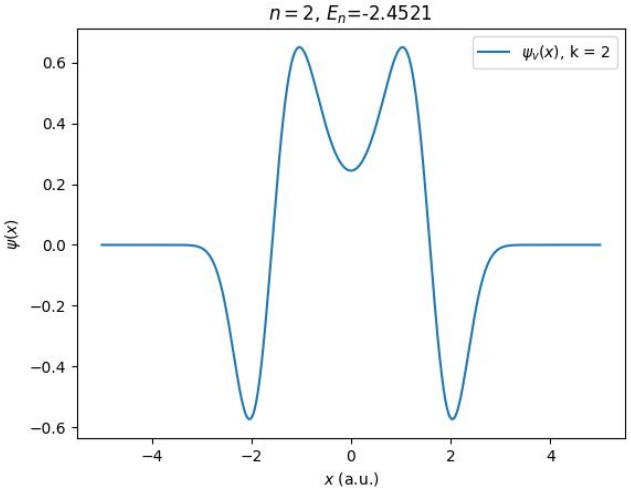
amb **b = 1** i **c = 6**.

Substituint els valors:

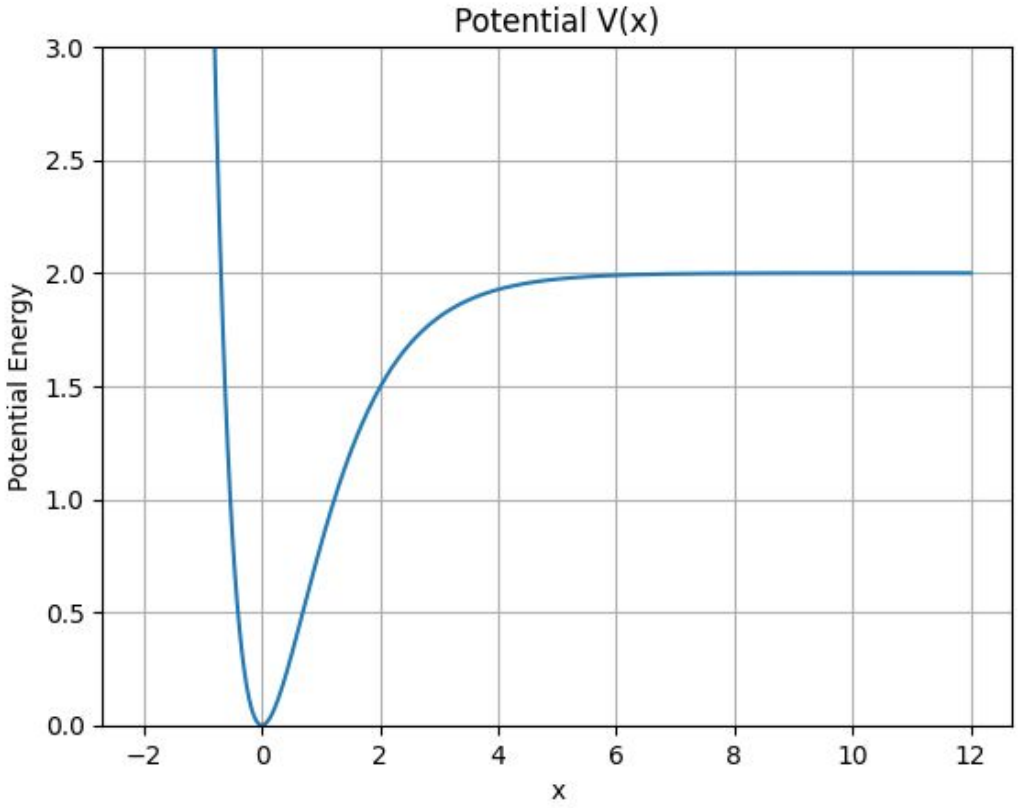
$\mathbf{V(x) = x^4 - 6x^2}$.

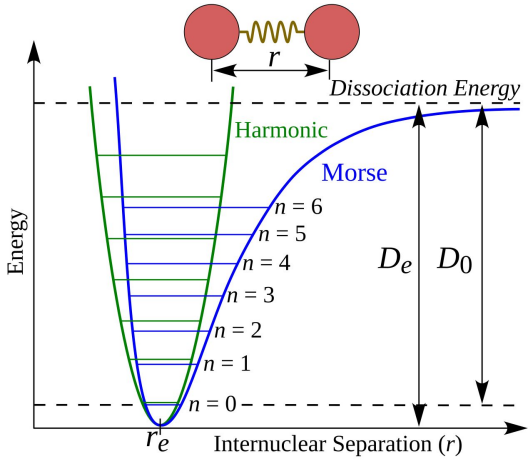

$n = 0$, E_n=-6.6430
$\psi(x)$
x (a.u.)
$\psi_v(x)$, k = 0


$n = 1$, E_n=-6.6409
$\psi(x)$
x (a.u.)
$\psi_v(x)$, k = 1


$n = 2$, E_n=-2.4521
$\psi(x)$
x (a.u.)
$\psi_v(x)$, k = 2

Morse Potential



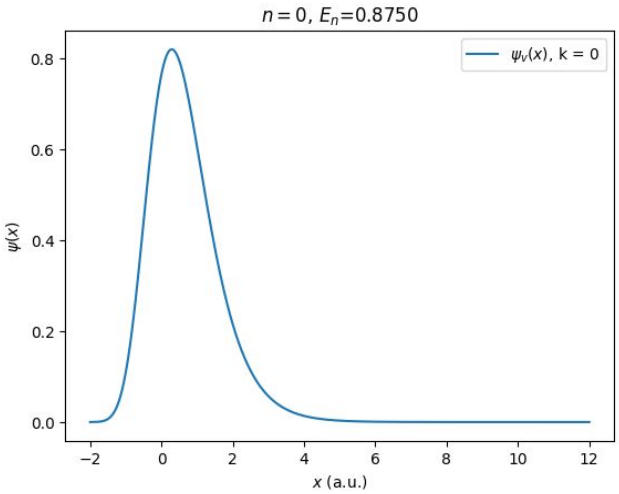


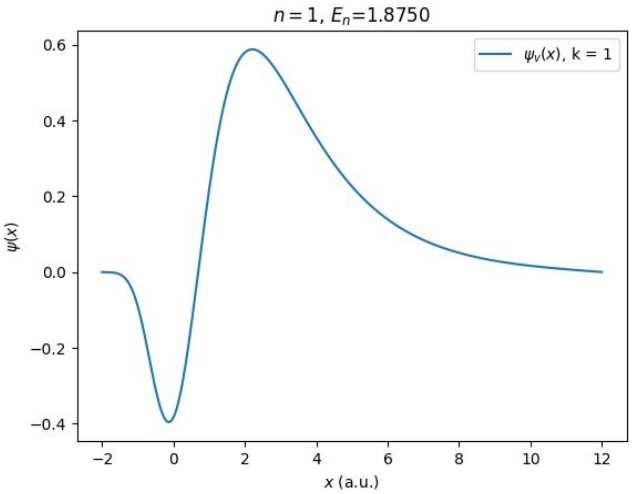
$$V(x) = D \cdot [1 - \exp(-B \cdot (x - x_e))]^2,$$

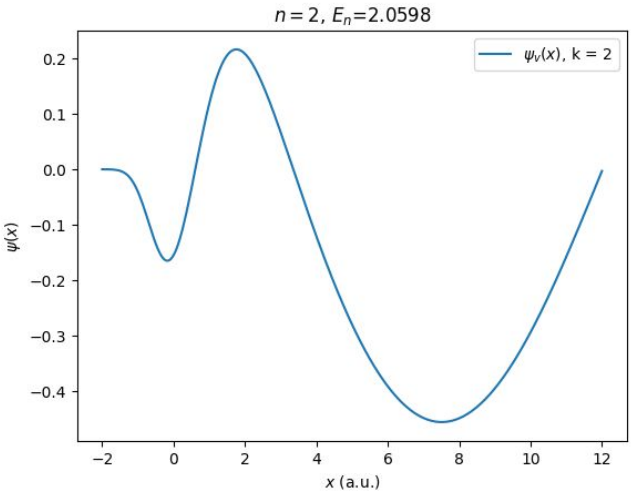
amb $D = 2$, $B = 1$ i $x_e = 0$.

Substituint els valors:

$$V(x) = 2 \cdot [1 - \exp(-x)]^2.$$


$n = 0$, E_n=0.8750
$\psi_v(x)$, k = 0
$\psi(x)$
x (a.u.)


$n = 1$, E_n=1.8750
$\psi_v(x)$, k = 1
$\psi(x)$
x (a.u.)


$n = 2$, E_n=2.0598
$\psi_v(x)$, k = 2
$\psi(x)$
x (a.u.)